

Experiment 19: Naive Bayes Classification for Bank Loan Prediction

Aim

To implement **Naive Bayes classification** for predicting whether a bank loan will be approved or not.

Python Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

data = {
    'Gender': ['Male', 'Female', 'Male', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male'],
    'Married': ['Yes', 'No', 'Yes', 'Yes', 'No', 'Yes', 'No', 'Yes', 'Yes', 'No'],
    'Education': ['Graduate', 'Graduate', 'Not Graduate', 'Graduate', 'Graduate',
                  'Not Graduate', 'Graduate', 'Graduate', 'Not Graduate', 'Graduate'],
    'ApplicantIncome': [5000, 3000, 2500, 6000, 3500, 2000, 4500, 8000, 2200, 7000],
    'LoanAmount': [150, 100, 90, 200, 120, 80, 130, 250, 85, 220],
    'Credit_History': [1, 1, 0, 1, 1, 0, 1, 1, 0, 1],
    'Loan_Status': ['Y', 'Y', 'N', 'Y', 'Y', 'N', 'Y', 'Y', 'N', 'Y']
}

df = pd.DataFrame(data)

print("Bank Loan Dataset:")
print(df)
le = LabelEncoder()

df['Gender'] = le.fit_transform(df['Gender'])
df['Married'] = le.fit_transform(df['Married'])
df['Education'] = le.fit_transform(df['Education'])
df['Loan_Status'] = le.fit_transform(df['Loan_Status'])
X = df.drop('Loan_Status', axis=1)
```

```

y = df['Loan_Status']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Naive Bayes model
model = GaussianNB()

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("\nPredicted Values:")
print(y_pred)

print("\nActual Values:")
print(y_test.values)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

new_applicant = [[1, 1, 0, 5500, 160, 1]]
prediction = model.predict(new_applicant)

if prediction[0] == 1:
    print("\nNew Applicant Loan Status: Approved")
else:
    print("\nNew Applicant Loan Status: Not Approved")

```

Sample Output

Bank Loan Dataset:

	Gender	Married	Education	ApplicantIncome	LoanAmount	Credit_History \
0	Male	Yes	Graduate	5000	150	1
1	Female	No	Graduate	3000	100	1
2	Male	Yes	Not Graduate	2500	90	0
3	Male	Yes	Graduate	6000	200	1
4	Female	No	Graduate	3500	120	1
5	Male	Yes	Not Graduate	2000	80	0
6	Female	No	Graduate	4500	130	1
7	Male	Yes	Graduate	8000	250	1
8	Female	Yes	Not Graduate	2200	85	0
9	Male	No	Graduate	7000	220	1

	Loan_ Status
0	Y
1	Y
2	N
3	Y
4	Y
5	N
6	Y
7	Y
8	N
9	Y

Predicted Values:

[1 1 1]

Actual Values:

[0 1 0]

Accuracy:

0.3333333333333333

Confusion Matrix:

[[0 2]

[0 1]]

Classification Report:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	2
1	0.33	1.00	0.50	1
accuracy			0.33	3
macro avg	0.17	0.50	0.25	3
weighted avg	0.11	0.33	0.17	3

New Applicant Loan Status: Approved